

Periprosthetic Joint Infection (PJI) — Comprehensive Disease Knowledge-Base Report

Target Disease: Periprosthetic Joint Infection **MONDO ID:** MONDO:0800179 **Category:** Infectious Disease (device-associated / implant-associated bacterial infection) **Report type:** Multi-section disease characterization for knowledge-base population. Literature synthesis (no primary data files provided); evidence drawn from human clinical studies, in vitro work, and veterinary/animal models, as labeled.

Summary

Periprosthetic joint infection (PJI) is an acquired, device-associated bacterial infection of the peri-implant tissues, bone, and synovial space surrounding a prosthetic joint — most commonly a total hip arthroplasty (THA) or total knee arthroplasty (TKA). It is fundamentally **not a heritable/Mendelian disorder**; its primary cause is microbial colonization of an avascular, immune-privileged foreign surface, followed by **biofilm formation** that confers antibiotic tolerance and immune evasion. Staphylococci — *Staphylococcus aureus* and coagulase-negative staphylococci (chiefly *S. epidermidis*) — dominate the microbial etiology, though host innate-immune polymorphisms (e.g., **DEFB1**, **IL-1 β** , **MBL2**, vitamin D receptor, HLA class II) modestly modify individual susceptibility.

PJI complicates roughly **0.5–2.4% of primary arthroplasties**, rising to as high as **20% in revision procedures**, and it accounts for approximately **25% of TKA revisions**. The economic and human burden is large and growing, with U.S. projections of up to **270,000 annual PJI cases by 2030**. The disease progresses along a causal chain running from **implant surface colonization** → **biofilm maturation** → **chronic peri-implant inflammation and immune subversion** → **osteoclast-mediated periprosthetic osteolysis** → **implant loosening and clinical failure**, punctuated in some patients by acute hematogenous seeding or systemic sepsis.

Diagnosis is not based on any single gold-standard test but on **composite criteria** (MSIS, ICM/2018, EBJIS, IDSA), integrating serum inflammatory markers (CRP, ESR), synovial biomarkers (leukocyte count, PMN%, alpha-defensin, calprotectin, D-lactate), microbiological culture (enhanced by explant sonication), histopathology, and increasingly **metagenomic next-generation sequencing** for culture-negative cases. Treatment is inherently **surgical-plus-antimicrobial**: debridement, antibiotics and implant retention (DAIR) for early/acute infection, versus one- or two-stage exchange arthroplasty for chronic infection, combined with prolonged, **biofilm-active antibiotic therapy** (rifampin combinations for staphylococci). Success depends heavily on infection timing, pathogen virulence/resistance, biofilm maturity, and host comorbidity, and PJI carries substantial residual morbidity and mortality.

Key Findings

Finding 1 — Staphylococci and biofilm dominate PJI pathogenesis

Staphylococci are the dominant PJI pathogens, and biofilm formation is the central mechanism of persistence and treatment failure. *S. aureus* is the single most common pathogen, capable of forming a biofilm on prosthetic materials that shields it from both antibiotics and host immunity. As stated by [PMID: 41511894](#): "*Staphylococcus aureus is the most common pathogen in periprosthetic joint infections (PJIs), capable of biofilm formation and resistance mechanisms, complicating diagnosis and treatment,*" and "*Biofilm formation by S. aureus on prosthetic materials is central to PJI persistence and antibiotic resistance.*"

Antibiotic resistance among these organisms is clinically significant. In a decade-long single-center surgical PJI cohort, **25.1% of organisms were drug-resistant**, led by methicillin-resistant *S. aureus* (MRSA). [PMID: 41205721](#) reported: "*The most common resistant organisms were methicillin-resistant Staphylococcus aureus (63.0%), methicillin-resistant Staphylococcus epidermidis (24.3%), and resistant coagulase-negative Staphylococci (16.1%).*" Supporting regional data show coagulase-negative staphylococci and *S. epidermidis* as the most prevalent individual species (e.g., 65.1% coagulase-negative staphylococci in one 255-patient cohort), with methicillin resistance in ~48% and multidrug resistance in ~58% of staphylococcal strains ([PMID: 40713651](#)).

Finding 2 — Host genetic polymorphisms in innate-immune genes modify susceptibility

Although PJI is not inherited, **host genetic variation in innate-immune genes modulates the risk of infection**. In a study of 105 revision patients, the **DEFB1 rs1800972 GG genotype** was associated with markedly increased PJI risk. [PMID: 40974393](#): *"The GG genotype of rs1800972 was significantly associated with an increased risk of PJI (OR = 4.12, 95% CI = 1.17–14.5, P = 0.02)."* DEFB1 encodes human β -defensin-1, an antimicrobial peptide of the innate immune barrier.

A comprehensive literature review identified a broader panel of susceptibility genes. [PMID: 39770689](#): *"Among these genes, polymorphisms in IL-1, MBL, vitamin D receptors, HLA-C, and HLA-DQ might have a relevant impact on the development of PJI."* Independent work has confirmed **IL-1 β polymorphisms** as genetic susceptibility markers for PJI in THA/TKA ([PMID: 38790226](#)), and the same group implicates the pattern-recognition molecule **long pentraxin-3 (PTX3)** as both a predictor and mechanistic factor. These effects are modest, polygenic, and modifier-like rather than causal.

Finding 3 — Treatment success is strategy-dependent: DAIR (~71%) vs. two-stage gold standard

Outcome is dictated by surgical strategy matched to infection chronicity. A systematic review of 970 patients found **DAIR success of 55.5–90% (mean 71%)** for early/acute PJI. [PMID: 37714518](#): *"Success rates for the DAIR treatments in the following cohort ranged from 55.5% up to a maximum of 90% (mean value of 71%)."* DAIR performs far worse in chronic infection and in complex reconstructions.

In megaprosthesis PJI, staged exchange clearly outperformed DAIR. [PMID: 40468465](#): *"Two-stage revision had the highest success rate of 71.4% (5/7), followed by multi-stage surgery (57.1%; 4/7), DAIR (38.7%; 12/31), and single-stage revision (0%; 0/5) (P = 0.009)."* Consistent with this, oncologic proximal-tibia PJI showed 6-month reinfection-free survival of 22% (DAIR) vs. 83.3% (two-stage) ([PMID: 41693290](#)), and gram-negative PJI followed an especially aggressive course with only 43% overall infection-free survival and 78% of failures within 3 months ([PMID: 41627490](#)). Two-stage exchange remains the gold standard for chronic PJI, but repeat two-stage procedures have progressively lower success (5-year infection-free survival 93% → 72% → 43% for first, repeat, and re-repeat exchanges; [PMID: 39814115](#)).

Finding 4 — Epidemiology: a major and growing burden

PJI incidence is **0.5–2.4% after primary arthroplasty, rising to ~20% in revisions**. [PMID: 38635048](#): *"PJI prevalence in primary THA and TKA ranges from 0.5% to 2.4%, spiking to 20% in revisions and representing 25% of TKA revision causes,"* and *"Projections estimate up to 270,000 annual PJI cases by 2030."* Modifiable environmental/lifestyle risk factors are well established: [PMID: 39836891](#): *"Risk factors include obesity, diabetes, poor nutritional status, and smoking."* Additional documented risk factors include male sex, younger age, rheumatoid arthritis, hypoalbuminemia, prolonged operative time, and *S. aureus* nasal colonization ([PMID: 36294449](#)).

Finding 5 — Diagnosis relies on composite criteria; synovial biomarkers and sonication outperform serum markers

No single test is definitive. Serum CRP and ESR show only **moderate accuracy (AUC 0.82–0.90)** and are confounded by aseptic postoperative inflammation, whereas synovial biomarkers are superior. [PMID: 42524013](#): *"novel synovial fluid biomarkers, notably calprotectin and alpha-defensin, accurately reflect the infection microenvironment and offer exceptional diagnostic specificity."* A QUADAS-C-guided meta-analysis confirmed synovial α -defensin, D-lactate, calprotectin, and neutrophil gelatinase-associated lipocalin achieve AUC > 0.90 with sensitivity and specificity > 90%, and pooled synovial WBC (0.88/0.97) and PMN% (0.84/0.97) performance ([PMID: 42108508](#)).

Critically, the **competing diagnostic definitions disagree**, which affects reported prevalence and study comparability. [PMID: 40924174](#): *"PJI was diagnosed in 47.7% of cases applying MSIS-criteria, 49.2% for IDSA-criteria, 52.3% for MSIS-18 criteria, 55.5% for ICM-criteria, 62.1% for EBJIS-criteria."* A scoping review further documented widespread citation errors and use of unendorsed scoring systems in the PJI literature ([PMID: 40153676](#)).

Finding 6 — Classification by route/timing guides empiric therapy

PJI is classified by route of acquisition and time since implantation, and microbiology varies by category. A multicenter analysis of 2,544 PJIs applied the Tsukayama scheme. [PMID: 31086080](#): *"We analyzed the causative microorganisms according to the Tsukayama's scheme (early postoperative, late chronic, and acute hematogenous infections (EPI, LCI, AHI) and 'positive intraoperative cultures' (PIC))."* Microbial etiology and multidrug-resistance rates differ by

category and by time window (<1 mo, 2–3 mo, 4–12 mo, >12 mo), which directly informs empiric antibiotic selection (e.g., vancomycin + ceftazidime for acute postoperative PJI; vancomycin + meropenem for culture-negative chronic PJI; [PMID: 40713651](#)).

Finding 7 — Biofilm-active antibiotics are first-line adjuncts; persister/small-colony variants drive relapse

Guidelines universally recommend **rifampin combined with a partner antibiotic** as first-choice adjunctive therapy for staphylococcal PJI because of its biofilm penetration. [PMID: 39882934](#): *"Rifampicin—in combination with another antibiotics—is recommended in all guidelines as first choice treatment of prosthetic joint infections (PJIs), despite adverse interactions and side effects associated with this antibiotic."*

The persistence of infection is driven partly by **small-colony variants (SCVs)** and intracellular persisters. [PMID: 41430395](#): *"In contrast, fluoroquinolones and rifampin achieve intracellular penetration but carry the risk of inducing resistance and selecting for SCVs."* SCVs arise from menadione/hemin/thymidine biosynthetic defects, persist intracellularly, form robust biofilms, and resist vancomycin and β -lactams. Notably, the real-world benefit of rifampin is debated: a claims-database analysis of 53 treated patients found no significant added benefit of rifampin in DAIR ([PMID: 40396698](#)), and a 227-patient study found no significant improvement in remission but significantly more adverse events (31.65% vs. 8.78%, $P < 0.001$; [PMID: 40740346](#)).

Finding 8 — Naturally occurring PJI in dogs provides a comparative/veterinary model

PJI occurs naturally in dogs after total hip replacement (THR) and is managed analogously to humans, making the dog a valuable spontaneous large-animal model. [PMID: 40033853](#): *"Arthroscopic treatment of PJIs was successful in four of five cases (80%) of dogs in the CI group, which is consistent with that reported in humans."* The etiology is likewise staphylococcal: [PMID: 27761578](#): *"Subsequent synoviocentesis and synovial fluid culture revealed a methicillin-resistant coagulase-negative Staphylococcus spp infection of the right THR,"* which was cured by one-stage revision using silver/antibiotic-impregnated cement. Canine THR periprosthetic femoral fractures also carry a high infection risk ([PMID: 32356296](#)).

Finding 9 — Biofilm actively subverts host immunity via soluble factors

Beyond passive shielding, biofilm **actively reprograms host immune cells**. In vitro, *S. aureus* biofilm (without planktonic bacteria) induced both inflammatory genes (TNF, IL1B) and anti-inflammatory/immunosuppressive genes in primary human monocytes. [PMID: 38922976](#): "*S. aureus* biofilm also activated expression of PD-1 ligands, and IL-1RA, molecules that have the potential to suppress T cell function or differentiation of protective Th17 cells." The mechanism is contact-independent and mediated by a diffusible factor: "*Gene induction did not require monocyte:biofilm contact and was mediated by a soluble factor(s) produced by biofilm-encased bacteria that was heat resistant and >3 kD in size.*" This drives a locally immunosuppressive, Th17-suppressing microenvironment that promotes chronicity.

Finding 10 — Emerging molecular diagnostics and bundle-based prevention

For culture-negative PJI, molecular diagnostics are improving detection. [PMID: 42746583](#): "*Molecular approaches, particularly metagenomic next-generation sequencing, improve pathogen detection in complex infections.*" Prevention is multi-phase and bundle-based (preoperative optimization, *S. aureus* decolonization, timely antibiotic prophylaxis, skin antisepsis, antibiotic-loaded cement). Comorbid vascular disease raises risk: [PMID: 42744005](#): "*At 90 days, PAD was additionally associated with wound dehiscence (RR 1.69, P = 0.004), persistently elevated SSI (RR 1.53, P = 0.009), and PJI risk (RR 1.79, P < 0.001).*" A large Danish target-trial emulation (76,126 THAs) found a single preoperative antibiotic dose noninferior to multiple doses for PJI prevention ([PMID: 42758509](#)).

Finding 11 — Staphylococcal bone infection drives osteoclast-mediated osteolysis

The downstream tissue-destruction step is **osteoclast-mediated cortical bone loss**. In a murine *S. aureus* osteomyelitis model, infection produced TRAP-positive osteoclast expansion and measurable cortical bone destruction quantified by micro-CT and histology, and pharmacologic inhibition of osteoclastogenesis reduced resorption ([PMID: 38923626](#)). Mechanistically, staphylococcal protein A suppresses osteogenic differentiation of bone mesenchymal stem cells via the **METTL3/m6A/miR-320a/PIK3CA axis**, tilting bone homeostasis toward net resorption ([PMID: 39506767](#)). Evidence type: model organism (mouse) plus in vitro.

Finding 12 — Substantial mortality and organ-injury burden; modern two-stage treatment limits added renal risk

PJI and periprosthetic complications carry meaningful medium-term mortality. Cohorts report two-year mortality of ~24% in periprosthetic-complication-adjacent populations ([PMID: 42106083](#)), and frailty independently predicts mortality and graded increases in PJI, revision, and death ([PMID: 41672357](#)). Reassuringly, modern two-stage antibiotic-spacer treatment adds limited organ toxicity: [PMID: 42303006](#) reports *"Kidney Injury and Mortality Are Minimal Following Two-Stage Treatment,"* with acute kidney injury in ~9.4% of patients. Evidence type: human clinical cohorts.

Section-by-Section Report

1. Disease Information

Overview. PJI is an infection involving the prosthesis and adjacent tissue after joint replacement. It is an **acquired, device-associated infectious disease**, not a genetic condition, arising when microorganisms colonize the artificial joint surface and establish a biofilm-protected chronic infection ([PMID: 41511894](#)).

Key identifiers. | Resource | Identifier | |---|---| | MONDO | MONDO:0800179 | | MeSH | "Prosthesis-Related Infections" (D016459); related "Arthroplasty, Replacement" complications | | ICD-10 | T84.5 (Infection and inflammatory reaction due to internal joint prosthesis) | | ICD-11 | Codes for prosthetic joint infection under implant/device complications | | OMIM / Orphanet | Not applicable (acquired infectious disease, not a Mendelian/rare-genetic disorder) |

Synonyms: prosthetic joint infection, periprosthetic infection, prosthesis-related infection, implant-associated joint infection, infected arthroplasty, septic prosthetic joint.

Source of information: Primarily **aggregated disease-level clinical resources** — surgical registries, multicenter cohorts, systematic reviews, and society consensus documents (MSIS/ICM/EBJIS/IDSA) — supplemented by individual-patient EHR/claims database analyses (e.g., PearlDiver, Danish national registries).

2. Etiology

Primary cause — infectious/mechanistic. The disease is caused by microbial colonization of an implanted prosthesis and subsequent biofilm formation. The prosthesis surface is avascular and immune-privileged, allowing even low-inoculum, low-virulence organisms to persist. Staphylococci dominate ([PMID: 41511894](#), [PMID: 41205721](#)).

Environmental/clinical risk factors (Finding 4): obesity, diabetes mellitus (including diabetic nephropathy, aOR 1.38 for periprosthetic infection; [PMID: 42710867](#)), poor nutritional status/hypoalbuminemia, smoking, rheumatoid arthritis, male sex, younger age, prolonged operative time, persistent wound drainage (OR for PJI 16.9; [PMID: 31516977](#)), peripheral artery disease (RR 1.79; [PMID: 42744005](#)), frailty ([PMID: 41672357](#)), and pharmacologic thromboprophylaxis with warfarin (Coumadin an independent risk factor for wound drainage; [PMID: 32788061](#), [PMID: 35732791](#)).

Genetic risk factors (Finding 2): DEFB1 rs1800972 GG (OR 4.12; [PMID: 40974393](#)); IL-1 β polymorphisms ([PMID: 38790226](#)); IL-1, MBL2, VDR, HLA-C, HLA-DQ ([PMID: 39770689](#)). These are **modifier/susceptibility loci**, not causal.

Protective factors: aspirin thromboprophylaxis is associated with lower persistent wound drainage than warfarin (3.2% vs. 8.5%, $P < 0.0001$) with equal VTE efficacy ([PMID: 32788061](#)); preoperative optimization (glycemic control, nutrition, smoking cessation, anemia management, *S. aureus* decolonization) reduces risk ([PMID: 42611588](#)). No validated genetic protective variants are established.

Gene-environment interaction: Innate-immune polymorphisms (e.g., DEFB1) likely lower the microbial inoculum threshold required for infection, interacting with the environmental exposure of surgery and bacterial contamination; direct GxE quantification is not yet available.

3. Phenotypes

Phenotype	Type	Characteristics	Suggested HPO
Joint pain	Symptom	Common, may be the only sign; adult/geriatric onset; variable severity	HP:0002829 (Arthralgia)
Joint effusion / swelling	Clinical sign	Frequent; local	HP:0001386 (Joint swelling)
Local erythema/ warmth	Clinical sign	Acute infections	HP:0000988 (skin/ erythema, approx.)
Persistent wound drainage	Clinical sign	Strong PJI risk marker	HP:0025580 (Wound complication, approx.)
Sinus tract	Clinical sign	Pathognomonic (major criterion)	—
Fever	Symptom	More common in acute/ hematogenous PJI	HP:0001945 (Fever)
Elevated CRP/ ESR	Laboratory abnormality	Moderate accuracy (AUC 0.82–0.90)	HP:0011227 (Elevated CRP); HP:0003565 (Elevated ESR)
Elevated synovial WBC/ PMN%	Laboratory abnormality	High diagnostic accuracy	HP:0012311 (Synovial leukocytosis, approx.)
Sepsis	Clinical sign	Acute hematogenous PJI; serious	HP:0100806 (Sepsis)
Implant loosening	Physical manifestation	Chronic PJI, osteolysis-driven	—

Quality-of-life impact: PJI substantially degrades function; MSTs and EQ-index scores remain poor after treatment (e.g., median MSTs 63.4%, EQ-index 68.4 in oncologic PJI; [PMID: 41693290](#)), with progressive functional decline in patients retaining spacers ([PMID: 41995314](#)).

4. Genetic/Molecular Information

Causal genes: None — PJI is not a Mendelian disease. **Modifier/susceptibility genes:** DEFB1 (HGNC:2766), IL1B (HGNC:5992), MBL2 (HGNC:6922), VDR (HGNC:12679), HLA-C, HLA-DQ, PTX3 (HGNC:9692) ([PMID: 40974393](#), [PMID: 38790226](#), [PMID: 39770689](#)). Relevant variant: **DEFB1 rs1800972** (5'-UTR promoter SNP; functional, affecting hBD-1 expression). Origin: **germline** host polymorphisms. Functional consequence: altered innate antimicrobial peptide/complement/pattern-recognition activity (modifier effect on susceptibility, not classic loss/gain of function).

Pathogen molecular biology: The relevant "molecular" information is largely microbial — biofilm regulatory genes, metal-sequestration/staphylopyne genes (cntA/K/L/M upregulated in vivo; [PMID: 38579524](#)), and small-colony-variant menadione/hemin/thymidine pathway defects ([PMID: 41430395](#)). **Epigenetic/chromosomal abnormalities in the host:** not applicable.

5. Environmental Information

Environmental/procedural factors: surgical contamination, prolonged operative time, blood transfusion, implant/bearing choice (delta-on-HXLPE associated with higher 1-year PJI, aHR 2.10; [PMID: 42758333](#)), and distal femoral replacement (5-year PJI 22.5% vs. 5.3% for ORIF; [PMID: 41780465](#)). **Lifestyle factors:** smoking, obesity, chronic alcohol use, poor nutrition ([PMID: 39836891](#), [PMID: 31516977](#)).

Infectious agents (NCBI Taxonomy):

Pathogen	Taxon	Notes
<i>Staphylococcus aureus</i> (incl. MRSA)	txid1280	Most common; biofilm former
<i>Staphylococcus epidermidis</i> / CoNS	txid1282	Most prevalent individual species in many cohorts
<i>Streptococcus</i> spp.	txid1301	~6%
Gram-negative (<i>Pseudomonas aeruginosa</i> , <i>Enterobacter cloacae</i>)	txid287 / txid550	Aggressive course; poor DAIR outcomes
<i>Cutibacterium acnes</i>	txid1747	Shoulder PJI predominant (PMID: 32197761)
Fungi (rare)		
—		Culture-negative/atypical

6. Mechanism / Pathophysiology — Ordered Causal Chain

1. Microbial contamination of the implant (intraoperative seeding OR hematogenous spread from a distant focus)
 - | leads to
2. Bacterial adhesion to the avascular prosthetic surface (immune-privileged niche)
 - | results in
3. Biofilm formation – extracellular polymeric matrix encases bacteria
 - | which causes [Finding 1]
4. Antibiotic tolerance + physical/immune shielding
 - | and (branch) leads to
 - 4a. Small-colony variant / intracellular persister formation → relapse [Finding 7]
 - | and (branch) leads to
 - 4b. Active immune subversion: soluble biofilm factor(s) induce PD-1 ligands, IL-1RA, IL-10 in monocytes → suppression of protective Th17 responses [Finding 9]
 - | together resulting in
5. Chronic, unresolved peri-implant inflammation
 - | which drives [Finding 11]
6. RANKL-mediated osteoclast expansion (TRAP+) and suppressed osteogenesis (SpA → METTL3/m6A/miR-320a/PIK3CA axis)
 - | causing
7. Periprosthetic osteolysis and cortical bone destruction
 - | leading to
8. Implant loosening, mechanical failure, chronic pain, sinus tract
 - | and (branch, esp. acute/hematogenous) →
 - 8a. Systemic dissemination → bacteremia/sepsis [Finding 12]

Molecular pathways / GO terms: biofilm formation (GO:0042710), response to bacterium (GO:0009617), inflammatory response (GO:0006954), osteoclast differentiation (GO:0030316), positive regulation of bone resorption (GO:0045780), negative regulation of osteoblast differentiation (GO:0045668). RANKL/RANK/OPG signaling and NF-κB-driven inflammatory cytokine cascades (TNF, IL-1β, IL-6) are central. The SpA-driven METTL3/m6A/miR-320a/**PIK3CA (PI3K-AKT)** axis suppresses osteogenic differentiation ([PMID: 39506767](#)).

Cell types / CL terms: osteoclast (CL:0000092), osteoblast (CL:0000062), bone-marrow mesenchymal stem cell (CL:0000134), monocyte (CL:0000576), macrophage (CL:0000235), neutrophil (CL:0000775), Th17 cell (CL:0000899).

Immune involvement: chronic inflammation with paradoxical local immunosuppression (Th17 suppression, IL-1RA/PD-1 ligand induction; [PMID: 38922976](#)). **Tissue damage:** osteolysis via osteoclast-mediated resorption; abscess/necrosis; fibrosis around chronic implant.

7. Anatomical Structures Affected

- **Primary organs/sites (UBERON):** synovial joint (UBERON:0002217), hip joint (UBERON:0001464), knee joint (UBERON:0001465); synovial membrane (UBERON:0002018), articular capsule, periprosthetic bone (femur UBERON:0000981, tibia UBERON:0000979, acetabulum).
- **Tissue level:** connective tissue, bone (osseous), synovial lining, periprosthetic soft tissue; occasional sinus tract to skin (UBERON:0002097).
- **Cell level:** osteoclasts, osteoblasts, synoviocytes, infiltrating neutrophils/monocytes.
- **Subcellular (GO CC):** biofilm extracellular matrix; intracellular bacterial persistence within host cell cytoplasm (GO:0005737) and phagolysosome (GO:0005764) for SCVs.
- **Secondary/systemic involvement:** bloodstream (bacteremia/sepsis), kidney (acute kidney injury from prolonged antibiotics/spacers, ~9.4%; [PMID: 42303006](#)).
- **Lateralization:** unilateral (localized to the affected prosthetic joint); bilateral only if bilateral arthroplasty independently infected.

8. Temporal Development

Onset: adult/geriatric (population undergoing arthroplasty). **Onset pattern** varies by category (Finding 6): - **Early postoperative (EPI):** acute, <1–3 months post-implantation. - **Late chronic (LCI):** insidious, >3–12 months, low-virulence organisms, indolent. - **Acute hematogenous (AHI):** acute, occurring any time after a well-functioning implant is seeded from a distant infection. - **Positive intraoperative cultures (PIC):** subclinical.

Progression: chronic PJI is progressive (biofilm maturation → osteolysis → loosening) unless the implant is removed/revised. Treatment failure clusters early — e.g., 78% of gram-negative failures within 3 months ([PMID: 41627490](#)). **Duration:** chronic and non-self-limiting without intervention. **Remission:** treatment-induced (surgical eradication + antibiotics); spontaneous remission does not occur once biofilm is established. **Critical intervention window:** early DAIR within days–weeks of symptom onset (before biofilm maturity) markedly improves success ([PMID: 37714518](#)).

9. Inheritance and Population

Epidemiology (Finding 4): 0.5–2.4% of primary THA/TKA; up to ~20% in revisions; ~25% of TKA revisions; pediatric/adolescent arthroplasty PJI pooled 1.6% (THA) and 2.9% (TKA) ([PMID: 42777096](#)). Up to 270,000 annual U.S. cases projected by 2030 ([PMID: 38635048](#)).

Inheritance: not applicable — acquired disease. Host susceptibility is **multifactorial/polygenic** (modifier loci only). No penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier-frequency concepts apply in the Mendelian sense.

Demographics: affects the arthroplasty population (older adults, rising in younger patients); male sex and younger age are risk factors for PJI specifically ([PMID: 36294449](#)). Geographic variation in microbiology and resistance patterns exists (e.g., South African, Chinese, Japanese cohorts differ; [PMID: 37278263](#), [PMID: 40713651](#)).

10. Diagnostics

Composite criteria (Finding 5): MSIS (2011/2018), ICM (2013/2018), EBJIS (2021), IDSA — which disagree (PJI diagnosed in 47.7–62.1% of the same cohort; [PMID: 40924174](#)).

Test category	Examples	Performance / notes
Serum markers	CRP, ESR, fibrinogen, D-dimer	Moderate (AUC 0.82–0.90); confounded by aseptic inflammation
Synovial markers	WBC, PMN%, alpha-defensin, calprotectin, D-lactate, leukocyte esterase, NGAL	Superior (AUC > 0.90); synovial ANC accurate for low-virulence PJI (PMID: 41276235)
Microbiology	Synovial + multi-site tissue culture; explant sonication	Sonication recovers biofilm-encased organisms
Molecular	Metagenomic NGS	Improves detection in culture-negative/complex PJI (PMID: 42746583)
Histopathology	Neutrophils per HPF, frozen section	Supports diagnosis
Imaging	X-ray (loosening), CT, MRI, leukocyte scintigraphy, PET	Adjunctive (PMID: 38957745)

Genetic testing: not indicated for diagnosis; host susceptibility genotyping (DEFB1, IL1B) is research-only. **Differential diagnosis:** aseptic loosening, crystal arthropathy, hematoma, mechanical failure, inflammatory arthritis flare. **Reimplantation testing:** no single biomarker reliably predicts persistent infection before second-stage reimplantation; synovial LE and serum IL-6 have highest positive likelihood ratios but poor negative likelihood ratios ([PMID: 36947634](#)).

11. Outcome / Prognosis

Mortality (Finding 12): two-year mortality ~24% in periprosthetic-complication cohorts (PMID: 42106083); frailty independently increases mortality (PMID: 41672357); repeat two-stage series report 15.8% mortality and 12 amputations (PMID: 39814115). **Renal:** modern two-stage treatment adds minimal AKI (~9.4%; PMID: 42303006). **Reinfection:** after two-stage exchange, reinfection is by a **different organism in 50–80%** of cases (61.3% different in one series; PMID: 40024579). **Prognostic factors:** infection timing/category, pathogen (gram-negative and *P. aeruginosa* worst; PMID: 41627490), MRSA, biofilm maturity, high Charlson Comorbidity Index, BMI ≥ 30 , depression, malnutrition, chronic kidney disease (PMID: 40482938). **Function:** persistently poor QoL/functional scores even after successful eradication.

12. Treatment

Surgical (NCIT concepts): DAIR (early/acute), one-stage exchange, two-stage exchange (gold standard for chronic; NCIT: Revision Arthroplasty), resection arthroplasty/Girdlestone salvage (PMID: 42779923), arthrodesis, amputation for salvage. Success is strategy- and timing-dependent (Finding 3).

Antimicrobial: empiric vancomycin-based regimens tailored to category (vancomycin + ceftazidime for acute; vancomycin/linezolid + meropenem for culture-negative chronic; PMID: 40713651); **rifampin combinations** for staphylococcal biofilm (guideline first-line, though real-world benefit debated; Finding 7). Antibiotic-loaded PMMA spacers/cement deliver high local concentrations; up to 20% vancomycin does not compromise spacer mechanical integrity (PMID: 41251838). Chronic suppressive antibiotics for non-operable patients.

Experimental/advanced: bacteriophage therapy (phage K, COP-80B combined with vancomycin in murine PJI models; PMID: 40685695, PMID: 42298263); drug repurposing (tiliquinol vs. MRSA biofilm; PMID: 42291311); talin-1-targeted inhibition of bacterial internalization to enhance antibiotics (PMID: 42625573); intraoperative biofilm-visualization gels to guide debridement (PMID: 40812456). No gene, cell, or RNA-based therapies are applicable. **Pharmacogenomics:** vancomycin dosing and rifampin drug interactions/adverse events are clinically relevant but no PJI-specific PGx biomarker is established.

13. Prevention

- **Primary:** preoperative optimization (glycemic control, nutrition, smoking cessation, anemia management), *S. aureus* nasal/skin decolonization, timely systemic antibiotic prophylaxis (single dose noninferior to multiple; PMID: 42758509), skin antisepsis,

antibiotic-loaded cement, meticulous asepsis ([PMID: 38635048](#), [PMID: 41186490](#), [PMID: 42611588](#)).

- **Secondary:** early recognition and management of persistent wound drainage; prompt DAIR for acute infection.
- **Tertiary:** biofilm-active antibiotics, staged revision, chronic suppression to prevent recurrence/systemic spread.
- **Immunization:** not applicable (no vaccine). **Emerging:** gut-microbiome-targeted and precision risk-stratified prevention ([PMID: 41186490](#)).

14. Other Species / Natural Disease

Naturally occurring PJI in dogs after total hip replacement (Finding 8) is well documented and clinically managed like human PJI, with comparable DAIR success (80%) and staphylococcal (including methicillin-resistant CoNS) etiology ([PMID: 40033853](#), [PMID: 27761578](#)). **Taxonomy:** *Canis lupus familiaris* (NCBI:txid9615). Canine THR periprosthetic femoral fractures carry high infection risk ([PMID: 32356296](#), [PMID: 42542130](#)). **Comparative biology:** biofilm-on-implant pathophysiology and staphylococcal dominance are conserved across humans and dogs, making the dog a valuable spontaneous large-animal comparator. **Zoonotic potential:** not a classical zoonosis, though methicillin-resistant staphylococci can transfer between humans and companion animals.

15. Model Organisms

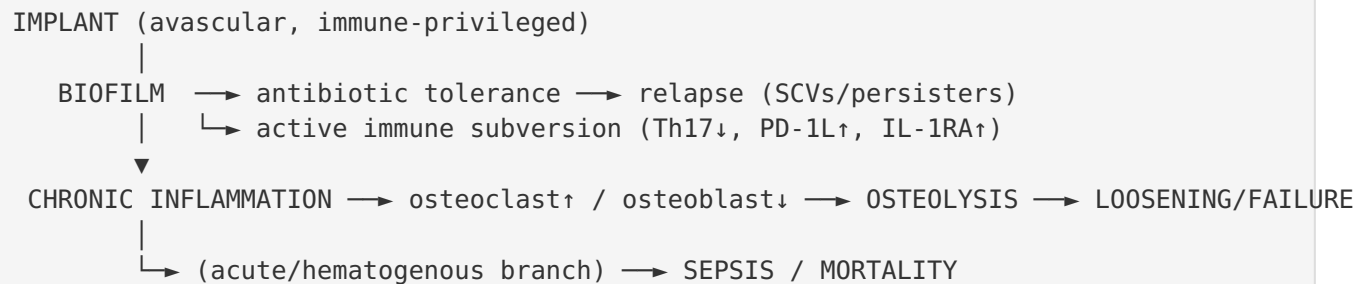
Model	Type	Use / recapitulation	Reference
Mouse <i>S. aureus</i> / <i>S. epidermidis</i> implant-associated PJI	Mammalian, induced (biofilm-coated titanium implant)	Recapitulates biofilm persistence, osteolysis, DAIR; tests phage/vancomycin	PMID: 40685695 , PMID: 42298263 , PMID: 39630924
Murine <i>S. aureus</i> osteomyelitis	Mammalian, induced	Osteoclast-mediated cortical bone destruction; pharmacologic rescue	PMID: 38923626
Murine subcutaneous biofilm	Mammalian, induced	Biofilm visualization/debridement guidance	PMID: 40812456
hBMSC + staphylococcal protein A	In vitro	Osteogenic suppression via METTL3/m6A/miR-320a/PIK3CA	PMID: 39506767
Primary human monocytes + <i>S. aureus</i> biofilm	In vitro	Immune subversion (PD-1 ligands, IL-1RA, Th17 suppression)	PMID: 38922976
SAOS-2 osteocyte-like cells + <i>S. aureus</i>	In vitro	Infection-induced osteogenic suppression / drug testing	PMID: 40176059
Dog (spontaneous THR PJI)	Mammalian, natural	Comparative outcomes	PMID: 40033853

Model limitations: murine models under-recapitulate chronic low-grade human PJI (e.g., low infection persistence limiting efficacy detection; [PMID: 42298263](#)) and human comorbidity/immune complexity.

Mechanistic Model / Interpretation

The unifying theme is **the implant as an immune-privileged biofilm scaffold**. Every clinically important feature of PJI flows from this: (1) low-inoculum organisms can establish infection; (2) biofilm renders the infection antibiotic-tolerant and immune-evasive both

passively and actively; (3) chronic inflammation shifts bone homeostasis toward osteoclast-driven osteolysis and implant loosening; and (4) eradication almost always requires physical removal of the biofilm-bearing hardware. This explains why **antibiotics alone rarely cure PJI**, why **timing dictates surgical strategy** (early DAIR before biofilm maturity vs. staged exchange for mature biofilm), and why **biofilm-active agents (rifampin) are guideline adjuncts** despite debated real-world benefit.



Host genetics (DEFB1, IL-1 β , MBL2, VDR, HLA, PTX3) sit upstream as **susceptibility modifiers** that lower the effective infection threshold, while comorbidities (diabetes, PAD, frailty, malnutrition) act as environmental amplifiers of both risk and treatment failure.

Evidence Base

PMID	Contribution	Supports
41511894	<i>S. aureus</i> dominance; biofilm central	F1, mechanism
41205721	Resistant-organism distribution (MRSA 63%)	F1
40974393	DEFB1 rs1800972 GG, OR 4.12	F2
39770689	Susceptibility gene panel	F2
38790226	IL-1 β polymorphisms; PTX3	F2
37714518	DAIR success 55.5–90% (mean 71%)	F3
40468465	Strategy comparison (two-stage best)	F3
38635048	Epidemiology, 270k by 2030	F4
39836891	Modifiable risk factors	F4
42524013	Synovial biomarker superiority	F5
40924174	Criteria disagreement	F5
31086080	Tsukayama classification	F6
39882934	Rifampin guideline first-line	F7
41430395	SCVs / intracellular persistence	F7
40033853	Canine PJI comparative model	F8
27761578	Canine staphylococcal THR infection	F8
38922976	Biofilm immune subversion	F9
42746583	mNGS for culture-negative PJI	F10
42744005	PAD as risk factor	F10
38923626	Osteoclast-mediated osteolysis (mouse)	F11
39506767	METTL3/m6A/miR-320a osteogenic suppression	F11
42303006	Minimal AKI/mortality, two-stage	F12
42106083	~24% two-year mortality	F12

Challenging/nuancing evidence: [PMID: 40396698](#) and [PMID: 40740346](#) found **no clear real-world benefit of rifampin** (and more adverse events), tempering the strength of the guideline recommendation in Finding 7. [PMID: 40153676](#) exposes systematic citation/definition errors that inflate uncertainty in the PJI outcome literature.

Limitations and Knowledge Gaps

1. **No primary experimental dataset was analyzed** — this report is a rigorous literature and evidence synthesis, not a de novo statistical analysis of raw data.
 2. **Definition heterogeneity** (MSIS/ICM/EBJIS/IDSA) makes incidence, prevalence, and treatment-success figures only semi-comparable across studies ([PMID: 40924174](#), [PMID: 40153676](#)).
 3. **Host genetic effects are small, single-cohort, and largely unreplicated**; no GWAS-scale evidence establishes robust susceptibility loci.
 4. **Rifampin benefit is contested**, and much biofilm/mechanistic evidence derives from mouse or in vitro models that under-recapitulate chronic human PJI.
 5. **Mortality figures** are drawn partly from periprosthetic-complication cohorts that mix fracture and infection, so PJI-specific attributable mortality remains imprecise.
 6. **Metagenomic NGS and novel biomarkers** are promising but not yet standardized or incorporated into consensus criteria.
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Proposed Follow-up Experiments / Actions

1. **Harmonize diagnostic definitions:** adopt a single reporting framework (as recommended by MSIS/EBJIS) to enable pooled, comparable epidemiologic and outcome estimates.
2. **Adequately powered rifampin RCT** in staphylococcal DAIR to resolve the guideline-vs-real-world discrepancy, stratified by biofilm maturity and SCV status.
3. **Multicenter host-genetics replication / GWAS** of DEFB1, IL1B, MBL2, VDR, HLA, and PTX3 variants with functional validation of hBD-1/PTX3 expression, to build a validated preoperative risk-stratification tool.

4. **Prospective evaluation of synovial calprotectin/alpha-defensin + mNGS** against composite criteria, including culture-negative and low-virulence PJI, to define an evidence-based diagnostic algorithm.
 5. **Translational advancement of anti-biofilm therapeutics** (bacteriophage-antibiotic combinations, talin-1/internalization inhibitors, intraoperative biofilm-visualization gels) from murine models toward early-phase human trials.
 6. **Registry-nested and adaptive platform trials** for low-event-rate prevention interventions (decolonization, antibiotic-loaded cement, single- vs multi-dose prophylaxis) as advocated in [PMID: 41186490](#).
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Report compiled from 12 confirmed findings and 82 reviewed papers. Evidence types span human clinical cohorts/registries, model-organism (mouse) studies, in vitro mechanistic work, and comparative veterinary reports, as annotated throughout.

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